

The QED Action

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August 2026

1 The Dirac Action

The Stern-Gerlach experiment brought about the need for a $1/2$ weighted representation of $SO(3)$. This was (as referenced in my other documents) found to be the double cover, $SU(2)$. This group transforms vectors mapped by σ into $\text{End}(S)$. To model an electron, we need to consider these objects between vectors and scalars, spinors. Dirac went about this without addressing it directly, he attempted to split the Klein-Gordon equation in two in order to make it quantum, in doing so, he obtained:

$$\left(i\sqrt{\partial_\mu\partial^\mu} - m\right)\psi = 0 \quad (1)$$

In order to make sense of the first term, he wrote:

$$\sqrt{\partial_\mu\partial^\mu} = \gamma_0\partial^0 - \gamma_1\partial^1 - \gamma_2\partial^2 - \gamma_3\partial^3 \quad (2)$$

To ensure that this expression squared back to what was expected, Dirac found the following properties:

$$\gamma_0^2 = +1 \quad \gamma_i^2 = -1 \quad \gamma_\mu\gamma_\nu = -\gamma_\nu\gamma_\mu \quad (3)$$

These form matrix representations of the basis vectors of the Clifford algebra $\text{Cl}(1,3)$. The Dirac equation is then:

$$(i\gamma_\mu\partial^\mu - m)\psi = 0 \quad (4)$$

Where ψ is a spinor valued field. Feynman introduced the "slash" notation:

$$\gamma_\mu A^\mu = \not{A} \quad (5)$$

We then write (4) as:

$$(i\not{\partial} - m)\psi = 0 \quad (6)$$

We can then reverse engineer a Lagrangian with the Euler-Lagrange equation in the adjoint field:

$$\frac{\partial\mathcal{L}}{\partial\bar{\psi}} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\bar{\psi})}\right) = 0 \quad (7)$$

A simple way to achieve this, with Lorentz invariance, is to multiply by the Dirac adjoint on the left:

$$\mathcal{L}_D = \bar{\psi}(i\not{\partial} - m)\psi \quad (8)$$

2 U(1) Gauge Theory

We may then construct a gauge theory using our U(1) symmetry. It is trivial to see that $\mathcal{L}_D$ has global U(1) symmetry:

$$\psi \rightarrow e^{ie\alpha} \psi \quad \bar{\psi} \rightarrow e^{-ie\alpha} \bar{\psi} \quad (9)$$

Suppose we localise this symmetry, we then gain extra terms in the derivative:

$$\partial_\mu \psi \rightarrow e^{ie\alpha(x)} (\partial_\mu \psi + ie(\partial_\mu \alpha) \psi) \quad (10)$$

We then introduce the covariant (or invariant) derivative to remove this extra term:

$$D_\mu = \partial_\mu + ieA_\mu \quad (11)$$

If and only if A_μ transforms as:

$$A_\mu \rightarrow A_\mu - \partial_\mu \alpha \quad (12)$$

Now, if we make this local U(1) transformation, the covariant derivative transforms as:

$$e^{ie\alpha(x)} (\partial_\mu \psi + ie(\partial_\mu \alpha) \psi + ie(A_\mu - \partial_\mu \alpha) \psi) = e^{ie\alpha(x)} (\partial_\mu \psi + ieA_\mu \psi) \quad (13)$$

We can see that the introduction of this gauge field A_μ has restored the symmetry that was broken by localising our U(1) element. Since we have introduced this new field, we need to bring in the field energy, along with the field strength tensor:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (14)$$

This may look familiar to you, this is the classical electromagnetic field strength tensor. Its field energy is:

$$\mathcal{L}_{EM} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \quad (15)$$

Therefore, our quantum electrodynamics Lagrangian is a sum of our gauge symmetric Dirac Lagrangian along with our electromagnetic Lagrangian:

$$\mathcal{L}_{QED} = \mathcal{L}_D + \mathcal{L}_{EM} \quad (16)$$

Which yields:

$$\mathcal{L}_{QED} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} (i\not{D} - m) \psi \quad (17)$$

The action is then:

$$S_{QED} = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} (i\not{D} - m) \psi \right) \quad (18)$$

Note that interactions are "hidden" in the covariant term. Expanding, we find it as:

$$S_{QED} = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} (i\not{\partial} - m) \psi - ej^\mu A_\mu \right) \quad (19)$$

Where j^μ is given as:

$$j^\mu = \bar{\psi} \gamma^\mu \psi \quad (20)$$